TechBridge Sol

Project Testing Policy

Effective Date: January 2026
Applies To: All Engineering, QA, and Development Staff

1. Purpose

The purpose of this Project Testing Policy is to define the standards, procedures, and responsibilities for testing all software projects at TechBridge Sol. This policy ensures that products meet functional, performance, security, and quality standards before deployment to production.

All engineers, QA staff, and project stakeholders are required to adhere to this policy.

2. Scope

This policy applies to all software development projects, including:

- Internal tools and platforms
- Client-facing SaaS applications
- APIs, microservices, and backend systems
- Mobile and web applications

- Integrations with third-party services

Testing must be conducted in **controlled environments** that mimic production as closely as possible.

3. Testing Principles

All projects must follow these principles:

- **Comprehensive:** Cover all functional and non-functional requirements
- **Early and Continuous:** Begin testing in early development stages and continue throughout the SDLC
- **Automated Where Possible:** Use automated tests for repetitive tasks, regression, and integration checks
- **Documented:** All test cases, results, and issues must be logged
- **Traceable:** Link test cases to project requirements and design specifications

4. Types of Testing

4.1 Unit Testing

- Tests individual components or functions
- Written and maintained by developers
- Automated and executed as part of CI pipelines

4.2 Integration Testing

- Tests interactions between modules or services
- Ensures that combined components work as intended
- Performed in a controlled staging environment

4.3 System Testing

- Validates the entire system against project requirements

- Covers functional, performance, security, and usability aspects

4.4 Regression Testing

- Ensures new changes do not break existing functionality
- Automated regression suites must be maintained and executed before releases

4.5 User Acceptance Testing (UAT)

- Conducted by end-users or project stakeholders
- Confirms the system meets business requirements and is ready for production

5. Testing Environment

- Maintain **separate environments** for development, testing, staging, and production
- Use **representative datasets** for realistic testing scenarios
- Ensure environments are secured and access is controlled
- Document all environment configurations and dependencies

6. Issue Reporting and Tracking

- All defects, bugs, and issues must be logged in the approved tracking system
- Include clear steps to reproduce, severity, screenshots, and affected modules
- Assign responsible personnel for resolution and retesting
- Track resolution status and closure for audit purposes

7. Test Case Management

- Maintain a **central repository** for all test cases
- Include metadata: requirement ID, test steps, expected results, and actual results
- Update test cases as project requirements evolve
- Conduct peer reviews of test cases for completeness and accuracy

8. Test Sign-Off

- Before deployment, a **Test Sign-Off** must be obtained from:
  - QA Lead or Engineer
  - Project Manager
  - Stakeholders (for UAT)
- Sign-off confirms that:
  - All planned tests have been executed
  - Critical defects are resolved or accepted
  - System meets project requirements and is ready for production

9. Policy Compliance

Non-compliance may result in:

- Deployment of unstable or defective software
- Delays in project timelines
- Escalation to engineering management for corrective action

Adherence to this policy ensures high-quality deliverables and minimizes production incidents.

10. Policy Review and Updates

This Project Testing Policy will be reviewed annually or when significant changes occur in technology, tools, or development practices. Engineers and QA staff are expected to comply with the latest version.